Kaleb Ben Naveed

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Education

- **The Hong Kong Polytechnic University** Hong Kong
  Bachelor of Engineering in Electronic & Information Engineering *Sep 2018 – May 2022 (Expected)*
  CGPA: 3.59/4.0
  ***Relevant Courses**: Deep Learning and Neural Networks, Perceptual Robotics, Computer Vision and Pattern Recognition, Probability and Statistics, Digital Signal Processing, Multimodal Learning, Computer Programming, Linear Algebra*
- **Pembroke College, University of Cambridge** Cambridge, England
  Exchange Semester at Pembroke and Kings College *July 2019 – August 2019*
  Final Grade: A (Cambridge "First Distinction")
  ***Relevant Courses**: Robotics (Planning & Control), Employability and Personal Branding, World Politics*

Publications

[1] **Kaleb Ben Naveed**, Zhiqian Qiao, and John M. Dolan, "Trajectory Planning for Autonomous Vehicles Using Hierarchical Reinforcement Learning," in Proceedings of IEEE Intelligent Transportation Systems Conference (ITSC '21), pp. 601 - 606, September 2021

[2] **Kaleb Ben Naveed**, Brady G. Moon, and Sebastian Scherer, "Informative and Fast Exploration Planning Using UAV for Reconnaissance Operations," in RISS Working Papers Journal (RISS '21), pp. 219 – 225, August 2021

Technical Skills

- **Languages:** Python, C++, MATLAB, C, R, LaTeX
- **Software:** ROS, Gazebo, AirSim, Unreal engine UE4, MATLAB & Simulink, Open AI gym, CARLA, SUMO, CUDA
- **Frameworks:** TensorFlow 1.x & 2.x, Keras, PyTorch, NumPy, Matplotlib, SciPy, OpenCV, Pandas
- **Electronics:** AVR microcontroller, Arduino, Soldering

Research Experience

- **Connected Autonomous Vehicles lab, The Hong Kong Polytechnic University** Hong Kong
  Research Assistant *Nov 2020 - Present*
  Supervisor: Dr. Edward Chung
  - Leading research on decision-theoretic planning using multiagent reinforcement learning
  - Developing algorithm for multi-agent on-ramp merging scenario for autonomous vehicles using PyTorch in SUMO
- **Air Lab, Carnegie Mellon University** Pittsburgh, PA
  Robotics Institute Summer Scholar (RISS) *May 2021 – Aug 2021*
  Supervisor: Dr. Sebastian Scherer
  [Paper][Poster][Video]
  - Proposed and developed Prioritized-FUEL (**F**ast **U**AV **E**xp**L**oration), a hierarchical approach for informative path planning
  - Proposed method showed **2 times faster Data Acquisition** with same exploration time compared to existent approach
  - Helped in Development of simulation environment using ROS, AirSim, Unreal Engine and Gazebo.
- **Argo AI, Carnegie Mellon University** Pittsburgh, PA
  Robotics Institute Summer Scholar (RISS) *April 2020 - Nov 2020*
  Supervisor: Dr. John M. Dolan
  [Paper][Poster][Video][Code]
  - Proposed and developed Robust-HRL [1], a Hierarchical Reinforcement Learning (HRL) based framework for decision making trajectory planning for Autonomous Vehicles.
  - Robust -HRL showed **at least 7 % higher success rate** than the existent state of the art RL and heuristic based approaches.
- **Engineering Laboratory, The University of Cambridge** Cambridge, England
  Summer Exchange Student at Pembroke College (PKP) *July 2019 - Aug 2019*
  Supervisor: Dr Gábor Csányi
  [Video]
  - Developed a Maze Solver line follower Robot which was successful to solve maze using sensors and pre-stored conditions.

- Added the ability to create the map of the path taken by robot, identify dead-ends and to do error correction.

Selected Projects

- **Final Year Project, The Hong Kong Polytechnic University** *Aug 2021 – Present*
  - Using machine learning techniques for radio map based UAV path planning with Disconnectivity constraint.
- **Capstone Project, Specialization in Robotics (MOOC), Coursera** *Sept 2021*
  - Used MATLAB to simulate the mobile inverted pendulum using PID controller and Extended Kalman Filter.
- **Autonomous Mini Robot Car as part of Integrated Project [Video]** *Sept 2020 – April 2021*
  - Developed robot car from scratch. The stages included PCB design and programming microcontroller and planning routine.
  - Was ranked in the highest category (top 10%) in class for having best performance metrics on all missions.
- **Technical AI Engineer, Aerovision Technologies, Hong Kong Science Park** *Dec 2020 – Jan 2021*
  - Improved accuracy by 25 % of the small-sized object detection framework using YOLO v4 for intrusion detection.
- **Student Assistant, PolyU School of Design** *May 2019 – Nov 2020*
  - Facilitated experiments and processed data for the project "Sizing and Comfort Study for Ear-related Product Design"

STEM Outreach & Leadership

- **Member and Contributor, CMU Robotics Outreach Working Group** Pittsburgh, PA
  Robotics Institute, Carnegie Mellon University *Sept 2021 - Present*
  - Designing and launching a website for students and teachers to learn about robotics. Website will include free learning resources about robotics, list of resources for different robotics opportunities, etc.
- **Founder and Leader of Reinforcement Learning Reading Group (RLRG)** Hong Kong
  Faculty of Engineering, The Hong Kong Polytechnic University *April 2021 - Present*
  - Founded RLRG to create a platform for students and faculty to learn and discuss RL and also to form possible collaborations. RLRG is first ever research reading group on campus
- **Assistant Lecturer for Cambridge GCE A levels** Lahore, Pakistan
  Science Department, Keynesian Institute of Management and Sciences *Sept 2017 - Feb 2018*
  - Responsible for teaching GCE A Level physics, chemistry laboratory experiments and programming problem sets.
  - Regularly helped students struggling with time management, and exam attempting techniques.

Talks & Presentations

Conference Paper Presentation, *IEEE Intelligent Transportation Systems Conference, Indianapolis* *Sept 2021*

RISS Research Poster Presentation, *Robotics Institute Summer Scholars Program, CMU* *Aug 2021*

Talk on Reinforcement Learning, *Teach Your Mates, Faculty of Engineering, HKPU* *April 2021*

RISS Research Poster Presentation, *Robotics Institute Summer Scholars Program, CMU* *Aug 2020*

Robotics Project Demonstration, *Pembroke College, University of Cambridge, UK* *Aug 2019*

Honor & Awards

- **Undergraduate Research and Innovation Scheme (URIS):** Awarded 5000 for my research project in connected autonomous vehicles lab. Awarded to almost 100 undergraduate students throughout university in June 2021
- **Undergraduate Summer Research Abroad Sponsorship (USRA):** Awarded 5000 USD to participate in Robotics Institute Summer Scholars Programme at Robotics Institute, Carnegie Mellon University in May 2021
- **Honored Global Student Ambassador:** Awarded highest level ambassadorship for organizing intercultural events
- **Oxbridge Programme Subsidy:** Awarded 6500 USD to participate in exchange semester at University of Cambridge
- **Deans Honor List 2018/19:** Included in Deans honor list for exceptional academic achievement.
- **Entry Scholarship by The Hong Kong Polytechnic University**: Full scholarship for 4-years BEng (Hons) in Electronic & Information Engineering. The scholarship covers full tuition fee, accommodation, and personal expenses.

Selected MOOC Courses

- **CS 285 Deep Reinforcement Learning**, *University of California, Berkeley, Online* *Aug 2021 – Present*
- **Probabilistic Graphical Model Specialization,** *Stanford University, Coursera* *May 2021 - Present*
- **Specialization in Robotics**, *University of Pennsylvania, Coursera* *June 2017 – Sept 2021*